C++ Notes: If Statements

(Conditions, logical operators, and comparisons)

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Learning Objectives

After this section, you will be able to:

- Use `if`, `else if`, and `else`
- Build conditions using boolean variables
- Combine conditions using `&&` (AND), `||` (OR), and `!` (NOT)
- Use comparison operators (`>`, `<`, `>=`, `<=`, `==`, `!=`)
- Create a `getMax` function using comparisons

Core Idea

Core idea

An `if` statement lets your program respond to different situations.
If a condition is **true**, run one block of code. If it is **false**, run another block (or skip it).

1. Basic `if` / `else`

Listing 1: Basic if/else with a boolean

```
bool isMale = true;

if (isMale) {
 cout << "You are a male" << endl;
} else {
 cout << "You are not male" << endl;
}
```

Example output (when `isMale = true`):

```
You are a male
```

Example output (when `isMale = false`):

```
You are not male
```

2. AND and OR

AND: both must be true (&&)

Listing 2: AND operator

```
bool isMale = true;
bool isTall = true;

if (isMale && isTall) {
  cout << "You are a tall male" << endl;
} else {
  cout << "You are not male or not tall" << endl;
}
```

OR: at least one must be true (||)

Listing 3: OR operator

```
bool isMale = true;
bool isTall = false;

if (isMale || isTall) {
  cout << "You are either male or tall (or both)" << endl;
} else {
  cout << "You are not male and not tall" << endl;
}
```

3. NOT (Negation) Operator: !

Listing 4: NOT operator

```
bool isTall = false;

if (!isTall) {
  cout << "You are not tall" << endl;
}
```

Mental model

`!condition` flips the boolean:

- if `condition` is `true`, then `!condition` is `false`
- if `condition` is `false`, then `!condition` is `true`

4. Full if / else if / else example

This version handles all combinations of `isMale` and `isTall`:

Listing 5: Multiple branches

```
bool isMale = true;
bool isTall = false;

if (isMale && isTall) {
  cout << "You are a tall male" << endl;
} else if (isMale && !isTall) {
  cout << "You are a short male" << endl;
} else if (!isMale && isTall) {
  cout << "You are tall, but not male" << endl;
} else {
  cout << "You are not male and not tall" << endl;
}
```

5. Comparisons inside if conditions

Comparisons

A comparison produces a boolean (`true` or `false`). That boolean can be used as the condition of an `if`.

Comparison operators (common ones)

- Greater than: `>`
- Less than: `<`
- Greater or equal: `>=`
- Less or equal: `<=`
- Equal (comparison): `==`
- Not equal: `!=`

Important

`=` is assignment. `==` is comparison.

6. Example Function: getMax of 2 numbers

Listing 6: Return the larger of two integers

```
int getMax(int num1, int num2) {
  int result;

  if (num1 > num2) {
    result = num1;
  } else {
    result = num2;
```

```
  }

  return result;
}
```

Test in main

Listing 7: Example run

```
int main() {
  cout << getMax(2, 5) << endl;
  return 0;
}
```

Output:

5

7. getMax of 3 numbers (more complex)

Listing 8: Return the largest of three integers

```
int getMax(int num1, int num2, int num3) {
  int result;

  if (num1 >= num2 && num1 >= num3) {
    result = num1;
  } else if (num2 >= num1 && num2 >= num3) {
    result = num2;
  } else {
    result = num3;
  }

  return result;
}
```

Test in main

Listing 9: Example tests

```
int main() {
  cout << getMax(1, 5, 10) << endl; // 10
  cout << getMax(10, 5, 1) << endl; // 10
  cout << getMax(5, 10, 1) << endl; // 10
  cout << getMax(200, 200, 1) << endl; // 200
  return 0;
}
```

Output:

```
10
10
10
200
```

Common Mistakes

Common mistakes

- Using `=` instead of `==` in conditions
- Forgetting parentheses: `if (condition)` not `if condition`
- Confusing `&&` with `||`
- Writing conditions that do not cover all cases (missing an else)

Self-Check

1. What is the difference between `&&` and `||`?
2. What does `!` do to a boolean condition?
3. Why is `==` different from `=`?
4. How does `getMax` decide which value to return?

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